

MELLE, Belgium

WMO: 064340

Lat: 50.9803N

Lon: 3.8158E

Elev: 49

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.2	25.5	14.2	11.3	23.6	19.6	14.7	28.3	25.5	50.8	23.9	49.9	6.4	60	0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	85.1	68.4	80.7	66.4	77.4	64.5	69.9	82.4	68.0	78.5	66.2	74.3	6.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	94.9	74.0	64.2	90.1	72.0	62.8	85.6	70.5	33.8	82.4	32.3	78.6	30.9	74.3	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.9	18.3	16.2	DB	16.8	93.4	7.3	4.2	11.5	96.4	7.3	98.9	3.1	101.2	-2.2	104.3	(4)
(5)			WB	16.2	73.9	7.2	1.2	11.0	74.8	6.7	75.5	2.7	76.1	-2.6	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	52.0	38.9	39.6	44.8	50.6	56.2	62.1	65.5	64.3	59.4	54.1	46.4	41.3		
	DBStd	11.16	7.29	7.17	6.26	6.41	6.47	5.35	5.36	4.52	4.78	6.27	6.35	7.19		
	HDD50	1353	346	292	180	70	13	0	0	0	1	31	143	277		
	HDD65	4921	809	710	627	434	283	120	57	67	180	341	558	735		
	CDD50	2084	2	2	18	87	206	364	481	443	282	157	35	7		
	CDD65	177	0	0	0	0	11	34	73	45	12	2	0	0		
	CDH74	1641	0	0	0	25	116	318	677	400	95	10	0	0		
	CDH80	438	0	0	0	0	13	77	209	121	18	0	0	0		
	WSAvg	7.8	8.9	9.2	8.4	7.6	7.7	7.5	6.7	6.4	6.4	7.4	8.4	9.5		
	PrecAvg	32.4	2.8	2.4	2.1	1.9	2.4	2.9	3.1	3.4	2.5	2.7	3.1	3.2		
(15) Precipitation	PrecMax	39.8	6.3	5.3	5.2	4.3	4.7	5.9	6.1	7.4	7.1	5.0	5.3	6.6		
	PrecMin	24.9	0.2	0.7	0.1	0.0	0.6	1.1	0.9	0.8	0.6	0.6	0.4	0.9		
	PrecStd	3.7	1.4	1.1	1.2	1.1	1.0	1.2	1.4	2.0	1.5	1.0	1.2	1.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.5	58.6	66.9	77.4	81.7	87.5	92.6	89.7	82.5	74.6	63.6	57.1		
		MCWB	52.8	49.0	54.0	59.0	64.9	70.2	69.9	70.3	67.8	64.5	58.9	53.2		
	2%	DB	53.2	53.4	62.2	72.8	77.4	81.8	85.5	83.2	76.2	70.0	60.7	55.1		
		MCWB	50.5	49.3	53.0	57.8	62.3	66.9	68.7	68.1	64.8	62.5	56.5	52.3		
	5%	DB	50.8	51.3	58.5	67.7	73.2	77.4	80.8	78.0	71.7	65.6	58.0	53.3		
		MCWB	48.3	47.9	51.0	55.7	60.7	64.6	66.4	65.7	62.7	59.6	54.6	50.7		
	10%	DB	48.4	49.3	54.8	62.1	68.4	73.3	77.0	74.0	68.4	62.9	55.6	51.0		
		MCWB	46.4	46.4	49.3	53.1	58.5	63.0	64.9	63.4	60.7	58.3	52.4	48.7		
	0.4%	WB	53.3	51.5	56.7	61.7	68.5	71.4	71.8	72.7	69.2	65.2	60.7	54.6		
		MCDB	54.8	55.3	64.3	74.1	78.7	83.2	86.0	84.5	80.4	73.4	62.9	55.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	50.6	49.8	53.8	58.4	65.1	68.5	69.9	69.1	66.4	62.6	57.0	52.7		
		MCDB	52.6	52.5	60.0	69.8	71.9	78.7	83.5	80.8	73.2	69.6	59.8	54.7		
	5%	WB	48.4	48.4	51.8	56.1	61.7	66.0	68.0	66.7	63.8	60.6	54.8	51.0		
		MCDB	50.4	51.1	57.6	65.9	70.6	75.0	78.9	75.5	69.4	64.9	57.9	53.1		
	10%	WB	46.5	46.6	49.8	54.1	59.4	63.7	66.0	65.0	62.1	58.9	52.7	48.9		
		MCDB	48.4	49.3	54.0	61.4	67.7	71.5	73.9	71.9	67.2	62.4	55.4	50.8		
	5% DB	MDBR	8.2	10.0	14.2	17.6	16.9	16.6	17.6	17.1	16.2	12.6	9.3	8.0		
		MCDBR	9.0	12.7	21.7	26.1	24.1	24.2	25.0	25.3	20.9	16.3	10.8	8.3		
		MCWBR	8.4	9.4	13.6	13.6	11.9	11.9	11.3	12.6	11.5	9.9	8.7	7.2		
		MCWBR	8.8	11.9	18.8	22.5	20.0	21.3	21.5	22.1	18.5	15.2	9.8	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.354	0.389	0.418	0.429	0.425	0.428	0.418	0.393	0.369	0.345	0.329		
		taud	2.371	2.339	2.238	2.184	2.202	2.233	2.230	2.276	2.335	2.392	2.375	2.341		
	Ebn at Noon		216	243	253	260	263	264	261	257	251	234	208	195		
		Edh at Noon	20	27	36	42	43	42	42	38	32	25	20	18		
(44) All-Sky Solar Radiation	RadAvg		248	467	846	1362	1601	1741	1647	1404	1049	609	291	192		
	RadStd		26	89	112	139	134	147	176	128	83	51	31	25		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (72 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page